Aakash Deep Yadav

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Skills

- **Languages**: Python, JavaScript, C, C++, Java, SQL
- **Frameworks/Libraries**: HTML, CSS, ReactJS, NodeJS, ExpressJS, Pandas, NumPy, Matplotlib, Scikit-Learn
- **Tools/Platforms**: MySQL, MongoDB, Oracle, Git, GitHub, AutoCAD, Illustrator
- **Soft Skills**: Problem-Solving, Adaptability, Analytical Thinking, Continuous Learning

Projects

Student Career Recommendation System | Github | Live *Oct'25 – Nov'25*

- Designed and deployed a full-stack ML-based career recommendation system that profiles students across **20+ attributes** (personality, skills, subject interests) and delivers **top-5 personalized career recommendations**.
- Implemented an unsupervised learning pipeline using KMeans++ and KMeans (Random), achieving a **Silhouette Score of ~0.82** and automatically selecting the optimal model based on 8+ clustering and performance metrics.
- Built a similarity-based recommendation engine using cosine similarity, processing **25+ career profiles** and generating ranked recommendations with **skill-gap analysis for 10 key skills**.
- Developed interactive analytics dashboards with 2D PCA and 3D UMAP visualizations, improving interpretability of high-dimensional ML outputs for end users.
- Architected a **3-service microservice system** (React, Node.js, FastAPI ML engine) with JWT-secured APIs, enabling real-time ML inference with sub-second latency.

Tech: React.js, Node.js, Express.js, MongoDB, Python, FastAPI, scikit-learn, REST APIs

Shared Expense Tracker | Github | Live *May'25 – Jun'25*

- Created a real-time shared expense tracking platform tailored for students living in PGs and apartments, enabling seamless group expense management and transparency.
- Secured user access using Firebase Authentication with role-based authorization, ensuring reliable data protection and controlled access across shared groups.
- Engineered a highly responsive live dashboard using React Hooks, efficient state management, and Firestore real-time listeners to deliver **sub-100 ms** expense update performance across users.
- Automated settlement logic to accurately calculate "who owes whom", significantly reducing manual reconciliation effort by **90%** and improving overall usability.

Tech: HTML, CSS, JavaScript, React.js, Express.js, Firebase

Training

Agentic AI Application Training – Programming Pathshala *Jun'25 – Aug'25*

- **About:** The intensive, hands-on training focused on designing, architecting, and building Agentic AI systems using modern LLM frameworks and multi-agent architectures for real-world applications.
- Built and deployed AI-driven applications integrating LLM APIs, backend services, and real-world business logic.
- Gained practical experience in LLM orchestration, autonomous agents, advanced prompt engineering, tool/function calling, memory handling, and workflow automation across diverse AI use cases.
- Strengthened understanding of AI system design, scalability, performance optimization, and production-ready AI application development.

Certificates

- Building Agentic AI Apps – **Programming Pathshala** | Link *Aug'25*
- Privacy and Security in Online Social Media – **NPTEL** | Link *Feb'24*
- Basic Python Towards AI/ML – **CSE Pathshala** | Link *Dec'23*

Achievement

Solved **100+ Data Structures & Algorithms problems** on LeetCode, strengthening problem-solving skills in arrays, strings, recursion, hashing, and basic algorithms. *Feb'26*

Education

- **Lovely Professional University** Punjab, India
  Bachelor of Technology - Computer Science and Engineering; ***CGPA: 8.19*** *Aug'23 – Present*
- **Tiny Tots Sr. Sec. School** Sahadatganj, Ayodhya
  Intermediate; ***Percentage: 82%*** *Apr'20 – Mar'22*
- **Tiny Tots Sr. Sec. School** Sahadatganj, Ayodhya
  Matriculation; ***Percentage: 83%*** *Apr'18 – Mar'20*